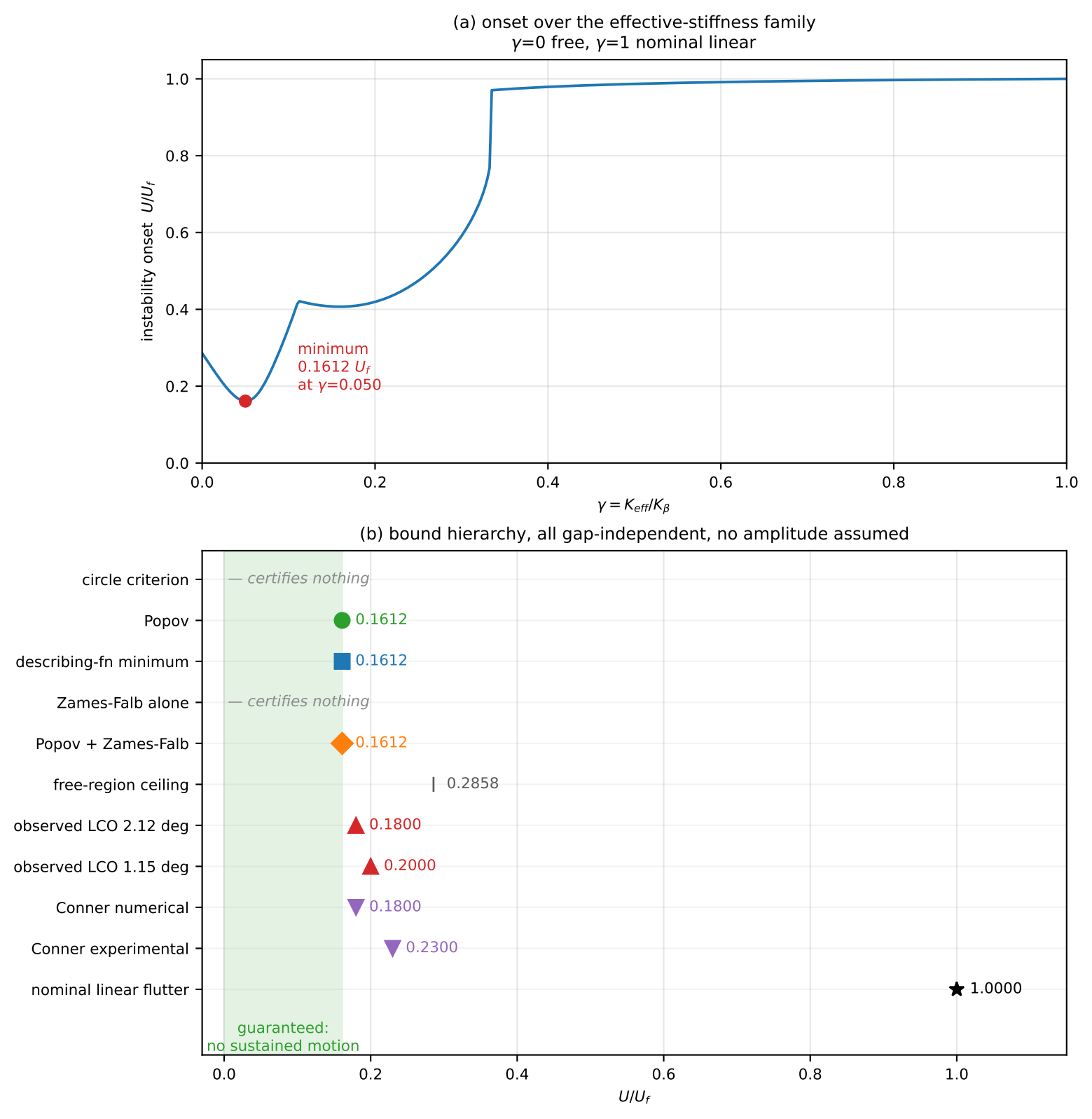
(a) onset over the effective-stiffness family
γ=0 free, γ=1 nominal linear
instability onset U/U_f
minimum
0.1612 U_f
at γ=0.050
γ = K_eff/K_β
(b) bound hierarchy, all gap-independent, no amplitude assumed
circle criterion
— certifies nothing
Popov
0.1612
describing-fn minimum
0.1612
Zames-Falb alone
— certifies nothing
Popov + Zames-Falb
0.1612
free-region ceiling
0.2858
observed LCO 2.12 deg
0.1800
observed LCO 1.15 deg
0.2000
Conner numerical
0.1800
Conner experimental
0.2300
nominal linear flutter
1.0000
guaranteed:
no sustained motion
U/U_f